Exercise Sheet #3: DC 2: Network Analysis

Discussion: Oct 29, 2025

Exercise 1: Method of Complete Tree

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Solution

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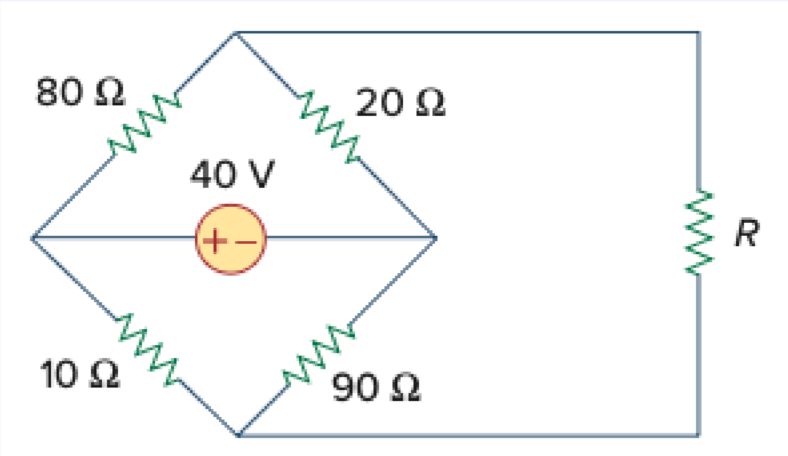
Exercise 2: Thévenin and Power Matching

The variable resistor R is adjusted until it absorbs the maximum power from the circuit.

a) Calculate the value of R for maximum power.

b) Determine the maximum power absorbed by R.



Solution

a) First, we calculate the Thévenin equivalent seen from the terminals of R.

Let node D (left of $40\,\mathrm{V}$ source) be at $40\,\mathrm{V}$, and node B (right of source) be at $0\,\mathrm{V}$.

For the upper node A:

$$\frac{V_A - 40}{80} + \frac{V_A - 0}{20} = 0 \Rightarrow 5V_A - 200 = 0 \Rightarrow V_A = 8\,\mathrm{V}$$

For the lower node C:

$$\frac{V_C - 40}{10} + \frac{V_C - 0}{90} = 0 \Rightarrow 9V_C - 360 = 0 \Rightarrow V_C = 36\,\mathrm{V}$$

Thus, the open-circuit (Thévenin) voltage is:

$$V_{\text{th}} = |V_A - V_C| = |8 - 36| = 28\,\mathrm{V}$$

Now we deactivate the $40\,\mathrm{V}$ source (replace by a short circuit):

$$R_A = (80\,\Omega \parallel 20\,\Omega) = \frac{80 \cdot 20}{80 + 20} = 16\,\Omega$$

$$R_C = (10\,\Omega \parallel 90\,\Omega) = \frac{10 \cdot 90}{10 + 90} = 9\,\Omega$$

Since R_A and R_C are in series,

$$R_{\text{th}} = R_A + R_C = 16 + 9 = 25\,\Omega.$$

For maximum power: $R = R_{\text{th}} = 25\,\Omega$

b) The circuit can be represented by a Thévenin voltage source V_{th} in series with a Thévenin resistance R_{th} and a load R_L.

$$I = \frac{V_{\text{th}}}{R_{\text{th}} + R_L}$$

Power in the load:

$$P = I^2 R_L = \frac{V_{\text{th}}^2 R_L}{(R_{\text{th}} + R_L)^2}$$

Substitute $R_L = R_{\text{th}}$ into the power equation:

$$P_{\max} = \frac{V_{\text{th}}^2 R_{\text{th}}}{(R_{\text{th}} + R_{\text{th}})^2} = \frac{V_{\text{th}}^2}{4R_{\text{th}}} = \frac{28^2}{4 \cdot 25} = \frac{784}{100} = 7.84\,\mathrm{W}$$

Exercise 3: Efficiency

If we want at least 75% efficiency, what fraction of the maximum load power do we get?

Solution

Efficiency η is defined as the ratio of power delivered to the load to the total power supplied by the source.
For a Thévenin source (V_{th}, R_{th}) with a load R_L, the efficiency is:

$$\eta = \frac{P_L}{P_L + P_{\text{th}}} = \frac{R_L}{R_L + R_{\text{th}}}$$

Given $\eta = 75\%$, solve for R_L:

$$0.75 = \frac{R_L}{R_L + R_{\text{th}}}$$

$$\Rightarrow R_L = 3R_{\text{th}}$$

At maximum power ($R_L = R_{\text{th}}$):

$$P_{L,\max} = \frac{V_{\text{th}}^2}{4R_{\text{th}}}.$$

Hence, the power ratio is:

$$\frac{P_L}{P_{L,\max}} = \frac{\dfrac{V_{\text{th}}^2 R_L}{(R_{\text{th}} + R_L)^2}}{\dfrac{V_{\text{th}}^2}{4R_{\text{th}}}} = 4R_{\text{th}} \cdot \frac{R_L}{(R_{\text{th}} + R_L)^2}.$$

Substitute $R_L = 3R_{\text{th}}$:

$$\frac{P_L}{P_{L,\max}} = 4R_{\text{th}} \cdot \frac{3R_{\text{th}}}{(R_{\text{th}} + 3R_{\text{th}})^2} = \frac{12R_{\text{th}}^2}{(4R_{\text{th}})^2} = 0.75.$$

Even when efficiency increases to 75%, the load still receives 75% of the maximum possible power. Thus, higher efficiency is achieved at the cost of only a moderate reduction in power delivered to the load.